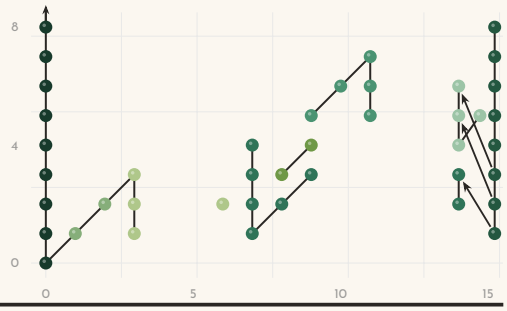
CHRISTIAN CARRICK

CURRICULUM VITAE



PERSONAL DETAILS

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RESEARCH Algebraic topology: equivariant, chromatic and motivic homotopy theory; synthetic spectra; higher real K-theories; stacks

POSITIONS AND EDUCATION

AUG 2025 - JUL 2027 **NSF Postdoctoral Fellow**
Mathematisches Institut, Universität Bonn
Supervisor: Markus Hausmann

JUN 2024 - JUL 2025 **NSF Postdoctoral Fellow (Guest Researcher)**
Utrecht Geometry Center, Universiteit Utrecht
Supervisor: Lennart Meier

JAN 2023 - MAY 2024 **Postdoctoral Researcher (Onderzoeker)**
Utrecht Geometry Center, Universiteit Utrecht
Supervisor: Lennart Meier

SEP 2022 - DEC 2022 **Trimester Program Participant**
Hausdorff Research Institute for Mathematics, Bonn
Spectral Methods in Algebra, Geometry, and Topology

SEP 2017 - JUN 2022 **Ph.D. Mathematics**
University of California, Los Angeles
Advisor: Michael A. Hill. Dissertation: *Stacks and Real-oriented homotopy theory*

2012 - 2016 **A.B. Mathematics**
Harvard University
Magna cum laude. GPA: 3.85. Advisor: Michael J. Hopkins. Senior thesis: *An elementary proof of Quillen's theorem for complex cobordism*

AWARDS AND HONORS

AWARDED APRIL 2024 **NSF Mathematical Sciences Postdoctoral Research Fellowship**
Three-year research fellowship from the National Science Foundation (DMS-2401918), held at Universiteit Utrecht and Universität Bonn.

JUNE 2022 **Pacific Journal of Mathematics Award**
Award for the three best dissertations in mathematics at UCLA; $3,500.

2021 - 2022 **UCLA Dissertation Year Fellowship**
Research fellowship, $20,000.

SUMMER 2016 **Harvard College Research Program Fellow**
Research fellowship, $3,000.

SPRING & FALL 2016 **Derek Bok Teaching Award**
For outstanding teaching as a course assistant at Harvard University.

SUMMER 2015 **PRISE Fellow**
Program for Research in Science and Engineering, Harvard University; research fellowship, $3,000.

FALL 2012 **Northern California Scholarship Foundation Winner**
Scholarship, $32,000.

PUBLICATIONS

[1] **The Real Brown–Peterson homology of $\Omega^\rho S^{\rho+1}$**
with Bert Guillou and Sarah Petersen
Submitted. arXiv:2512.18019

[2] **Slice spectral sequences through synthetic spectra**
Submitted. arXiv:2510.19501

[3] **On higher real K-theories and finite spectra**
with Mike Hill
Submitted. arXiv:2507.07051

[4] **On periodic families in the stable stems of height two**
with Jack Davies
Submitted. arXiv:2506.20507

[5] **The descent spectral sequence for topological modular forms**
with Jack Davies and Sven van Nigtevecht
Submitted. arXiv:2412.01640

[6] **Nonvanishing of products in v_2-periodic families at the prime 3**
with Jack Davies
Advances in Mathematics 502 (2026), 111183

[7] **Chromatic defect, Wood's theorem, and higher real K-theories**
Geometry & Topology 30(3) (2026), 1051–1108

[8] **A synthetic approach to detecting v_1-periodic families**
with Jack Davies
Transactions of the AMS 379(3) (2026), 1985–2031

[9] **On MU-homology of connective models of higher real K-theories**
with Mike Hill
Proceedings of the AMS, Series B 12 (2025), 172–186

[10] **Descent spectral sequences through synthetic spectra**
with Jack Davies and Sven van Nigtevecht
International Mathematics Research Notices 2025(15), rnaf216

[11] **The homological slice spectral sequence in motivic and Real bordism**
with Mike Hill and Doug Ravenel
Advances in Mathematics 458, Part A (2024), 109955

[12] **Cofreeness in Real bordism theory and the Segal conjecture**
Proceedings of the AMS 150(7) (2022), 3161–3175

[13] **Smashing localizations in equivariant stable homotopy**
Journal of Homotopy and Related Structures 17(3) (2022), 355–392

INVITED TALKS

10/2026 **eCHT Online Research Seminar**
Multiplication on connective higher real K-theories

1/2026 **Wuppertal Topology Seminar**
Higher real K-theories and finite spectra

11/2025 **Beijing Algebraic Topology Seminar**
Slice spectral sequences through synthetic spectra

11/2025 **Bonn Algebraic Topology Seminar**
Slice spectral sequences through synthetic spectra

10/2025 **Princeton Algebraic Topology Seminar**
Higher real K-theories and finite spectra

4/2025 **Utrecht–Nijmegen TopICS Seminar**
Higher real K-theories and finite spectra

11/2024 **University of Bochum Topology Seminar**
Chromatic defect, Wood's theorem, and higher real K-theories

3/2024 **New Mexico Topology Seminar**
Synthetic spectra and the image of J

11/2023 **eCHT Online Research Seminar**
Chromatic defect, Wood's theorem, and higher real K-theories

10/2023 **University of Groningen**
The Segal conjecture and $\mathbb{F}_1$

10/2023 **University of Bonn Topology Seminar**
Chromatic defect, Wood's theorem, and higher real K-theories

7/2023 **International Workshop on Algebraic Topology, Beijing**
Chromatic numbers and Real-oriented spectra

2/2023 **Utrecht Geometry Center Seminar**
The Segal conjecture and $\mathbb{F}_1$

4/2022 **Chico Topology Conference (contributed)**
A new proof of the Segal conjecture via Real bordism theory

3/2022 **UCSD Visiting Algebraic Topology Seminar**
The homology of $BP_{\mathbb{R}}\langle n\rangle$

11/2021 **Chicagoland Algebraic Topology Seminar (University of Chicago)**
The Segal conjecture and Real bordism theory

11/2021 **AMS Fall Western Sectional Meeting, Special Session on Equivariant and Motivic Homotopy Theory**
The Segal conjecture and Real bordism theory

10/2019 **UCLA Visiting Algebraic Topology Seminar**
Smashing localizations in equivariant stable homotopy

SERVICE

15-19 MAR 2027 **Workshop on Synthetic Spectra, Bergische Universität Wuppertal**
Co-organizer, with Sven van Nigtevecht and Fei Ren, of a week-long workshop on the theory and computational sides of synthetic spectra, with lecture series by Irakli Patchkoria and Dan Isaksen.

2023 - 2024 **TopICS Intercity Topology Seminar**
Organized a monthly invited seminar joint with Universiteit Utrecht, Radboud Universiteit, and Vrije Universiteit Amsterdam.

SINCE 2021 **`ssplot.netlify.app`**
Developed and maintain a website to display and manipulate spectral sequence diagrams automatically from Macaulay2 output.

WINTER & SPRING 2021 **UCLA Directed Research Program (DRP) Mentor**
Mentored undergraduates in chromatic homotopy theory.

2020 - 2021 **UCLA Participating Algebraic Topology Seminar**
Organized seminars on higher real K-theories (Fall 2020), stacks (Winter 2020), and motivic homotopy theory (Fall 2021).

2018 - 2019 **UCLA Juvitop Seminar**
Founded the seminar in Fall 2018; organized semesters on computations in chromatic homotopy theory (Fall 2018) and structured ring spectra (Winter 2019).

2015 - 2016 **Harvard College Tutoring Programs**
Tutored high school students in the Cambridge After School Program and the CHANCE tutoring program.

TEACHING AND ADVISING

2025 - PRESENT **Instructor** / Universität Bonn

- **Algebraic Topology I** (V4D1), Master's lecture course. Winter 2026/27.
- **Mini-course:** six-week Master's course on the Kervaire invariant one problem, with an accompanying YouTube series. 2/2026 – 3/2026.
- **Master's theses advised:**
  - Daniel Arone, *Vanishing in slice spectral sequences and nilpotence for the Real Brown–Peterson spectrum*, 2026.
  - Ben Steffan, *Smith ideals in higher algebra*, 2026.

2023 - 2025 **Instructor** / Universiteit Utrecht

- **Bachelor's thesis advised:** Chris Vos and Jacco Hijmans, joint thesis on the Adams spectral sequence and Ext over the Steenrod algebra; the students also wrote flexible, sophisticated Ext-resolution software. Spring 2024.
- **Master's thesis advised:** Luca Dal Forno, Day convolution for equivariant ∞-operads. Spring 2024.
- **Master's seminar:** Topological K-theory. Spring 2024.
- **Orientation to Mathematical Research:** Abelian and derived categories; advised four Master's students on a research project in homological algebra. Winter 2024.

2017 - 2022 **Teaching Assistant** / UCLA Department of Mathematics

- **Math 33A**, Linear Algebra: Winter 2021.
- **Math 31B**, Calculus II: Fall 2018, Winter 2019, Spring 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021.
- **Math 31A**, Calculus I: Fall 2018, Winter 2019, Spring 2019, Fall 2019, Fall 2020.
- **Math 32B**, Multivariable Calculus II: Spring 2018, Summer 2019, Winter 2020, Summer 2020.
- **Math 32A**, Multivariable Calculus I: Winter 2018.
- **Math 1A**, Precalculus: Fall 2017.

2016 **Course Assistant** / Harvard University, Department of Mathematics

- **Math 131**, Topology: Fall 2016. Derek Bok Teaching Award.
- **Math 101**, Sets, Groups, and Topology: Spring 2016 (Derek Bok Teaching Award) and Fall 2016.

REFERENCES

RESEARCH	Michael A. Hill	mahill@umn.edu
	Lennart Meier	f.l.m.meier@uu.nl
	Douglas Ravenel	dcravenel@gmail.com
TEACHING	Martha Contreras	marthac@seas.ucla.edu